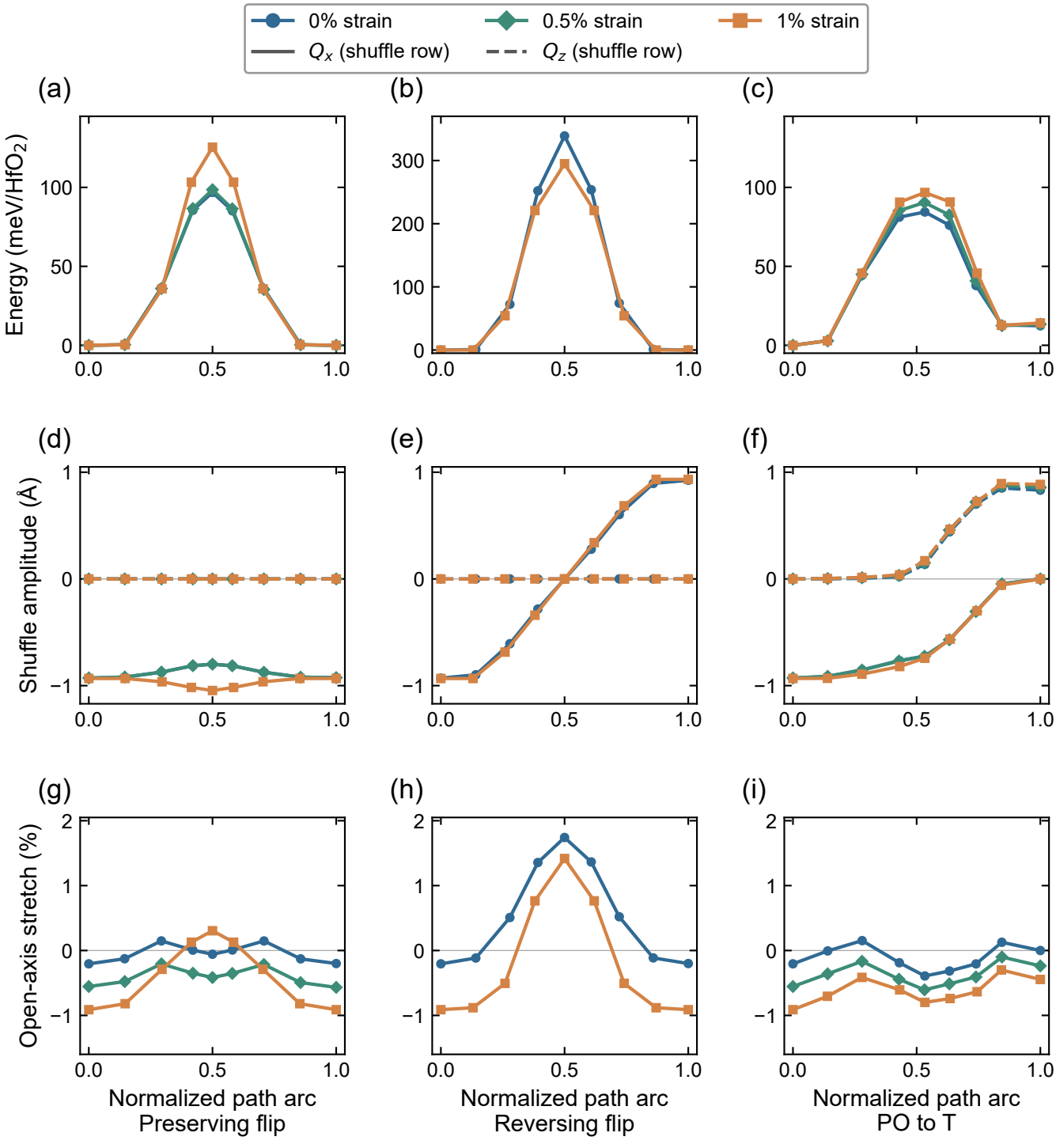
0% strain
0.5% strain
1% strain
Q_x (shuffle row)
Q_z (shuffle row)
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
Energy (meV/HfO$_2$)
Shuffle amplitude (Å)
Open-axis stretch (%)
Normalized path arc
Preserving flip
Normalized path arc
Reversing flip
Normalized path arc
PO to T